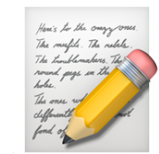


Upcoming + Reminders

Find all links on the course website: cogs9.github.io/fa26



Read: Reading(s) 3, *Data Organization*



Assignments (find links on course website):

- Reading quiz 1, pre-course survey, syllabus quiz (**DUE: TODAY**)
- Reading quiz 2 (**DUE: Wed week 3**) *All due at 11:59pm*

EXAM 1 is Monday, October 19

Group Project Guidelines

Groups of 4-5 students

1. Form your own group OR
2. Be matched with a group based on interests / work styles

- If you want to be matched OR want 1-3 more people in your group, please fill out the **group-finding survey** by Wednesday of week 3!
- We will match groups by the end of that week BY EMAIL (contact your group ASAP)

Group Project Checkpoint 1 is
DUE Wednesday of Week 5

Project group-finding survey | DUE: Wed week 3)

IF YOU ALREADY FOUND OR FORMED A PROJECT GROUP OF 4-5 STUDENTS, please do **NOT** fill out this form!! Your group may go ahead and start on [Group Project Checkpoint 1!](#) (No need to fill out this form)

Please only fill out this form if...

1. you want to be **assigned** to a group of students with similar interests (a great way to meet new friends!)
2. **OR** if you want to **ADD** 1-3 more people to your group! (only **ONE** person from the group needs to fill this out)

COGS 9 | Week 2

The background is a dark blue gradient. On the right side, there is a vertical column of binary code (0s and 1s) in a light blue color. A yellow sine wave curves across the middle of the image. In the foreground, there is a candlestick chart with red and green bars, and a yellow trend line that follows the general direction of the bars.

Review: Data Types
&
Obtaining Data

Brendan Tomoschuk

Data Types — Review

- Data values can be...
- **Quantitative:** a value that can be *objectively* measured or counted
 - **Continuous** (e.g., height, temperate, population)
 - **Discrete** (e.g., number of kids, score in a game)
- **Qualitative:** describes categories or qualities
 - **Categorical**
 - Nominal or *unordered* sets (e.g., colors, blood type, true/false)
 - Ordinal or *ordered* sets (e.g., low, medium, high)
 - Text, audio, images (in raw form)

Discussion (~7 mins)

With a group of 2-3:

Where and when do bike/scooter accidents most likely happen around campus, and what interventions would reduce them?

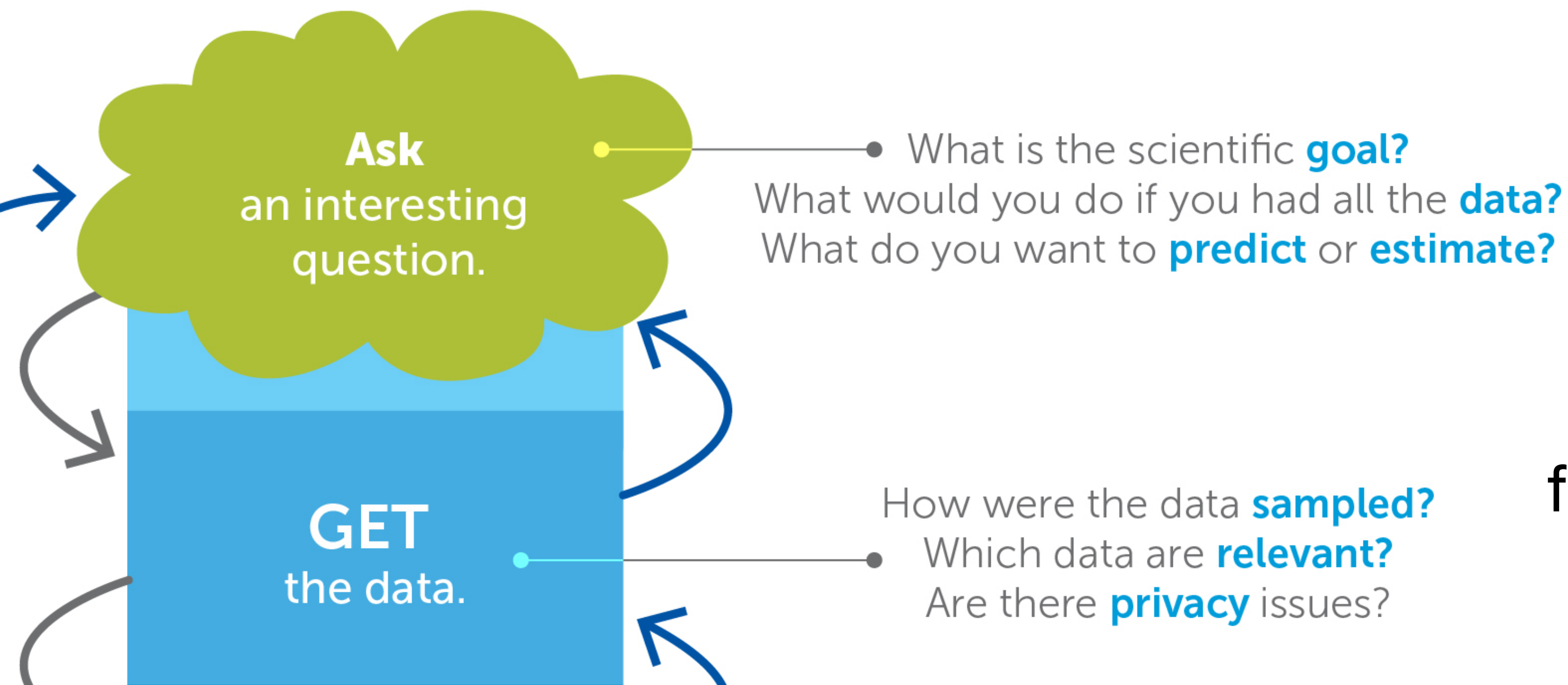
1. What is the goal(s) of this question?
(Describe, predict, test cause-and-effect, and decide)
2. What data you would need to collect or obtain to answer this question?
3. For each type of data, is it **quantitative** or **qualitative**? Is it **structured** or **unstructured**?

Takeaways

- Data can be classified into **quantitative** or **qualitative** data types
 - Different data types require different summaries / visualizations
- Data can be stored in a variety of formats ranging from **unstructured** to **structured**
- Let your data science question **guide your choice of data!**

Last week: Ethics, Privacy, & Data

- Consider ethical & privacy concerns throughout your data science project to prevent unintended harm
- Many different types of data (qualitative/quantitative, structured/unstructured) and many ways to store it



This week: how do we
get data & prepare it
for analysis and modeling?

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course code:
WSTAPX

What is a relational database?

- (A) A database that stores unstructured data
- (B) A database that organizes data into tables with rows and columns that can be linked
- (C) A database designed for storing images and videos
- (D) A database is a spreadsheet
- (E) I'm not sure / I'm confused

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course code:
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course code:

WSTAPX

**A credit card account number is a
_____ variable**

- (A) Quantitative, categorical
- (B) Quantitative, discrete
- (C) Qualitative, nominal
- (D) Qualitative, ordinal
- (E) I'm not sure / I'm confused

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course code:
WSTAPX

**A credit card account number is a
_____ variable**

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Learning Goals

1. Identify and select appropriate methods of obtaining data
2. Consider the broad ethical considerations in obtaining and using data (origin, privacy, consent, fairness, and appropriate use)

What are some ways we can get data?

(~2 min with your neighbor)

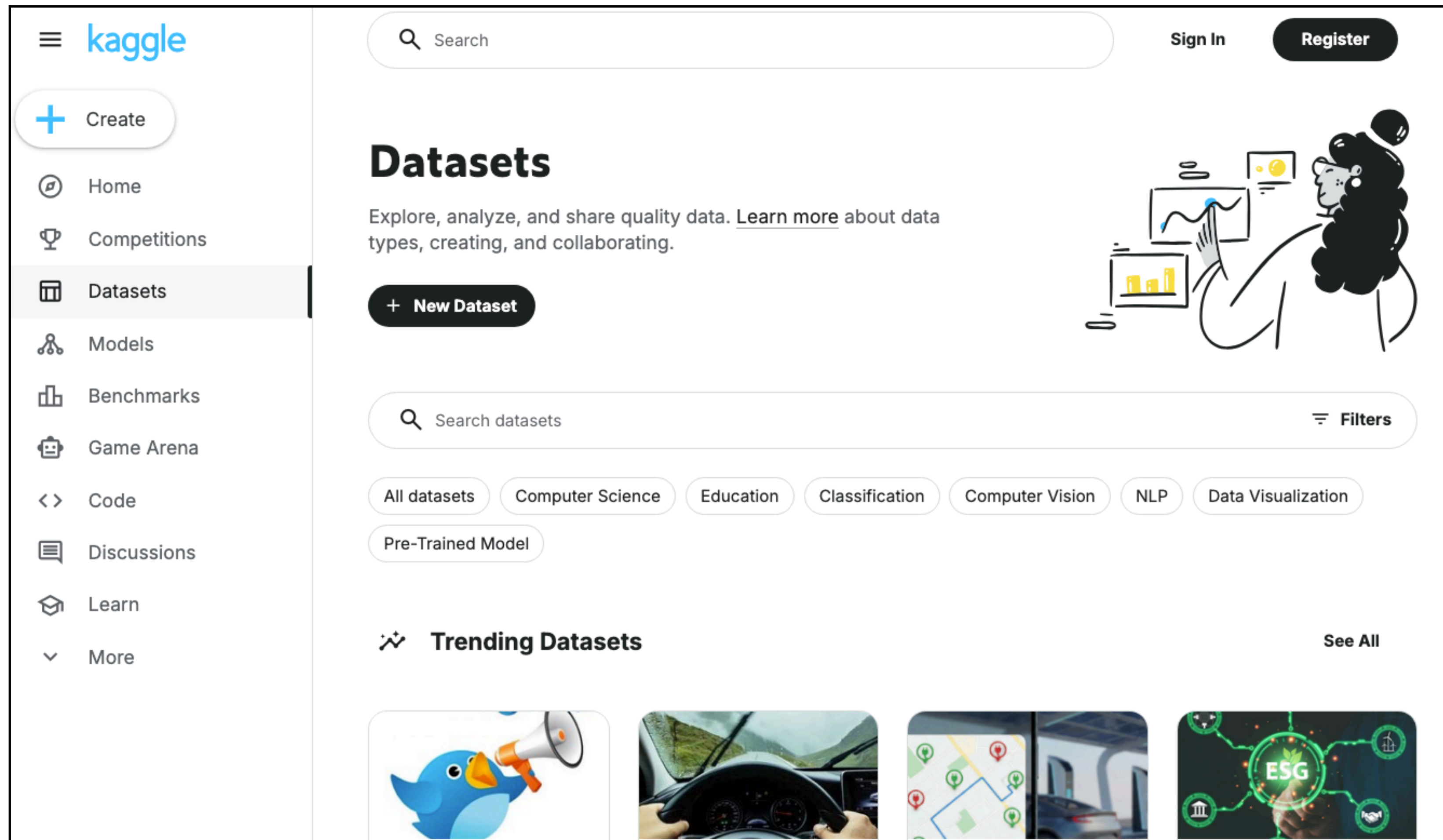
How to get data

1. Collect it yourself
2. Use public / open-source data
3. Buy it
4. Request it from an API
5. Scrape it from a website
6. Source it from your company

Primary Vs. Secondary Data Analysis	
Primary	Secondary
Data is collected first hand for a specific research aim.	Data was collected by someone else for a different purpose
Data is up-to-date as it is collected for the current research.	Data may be outdated and not specific to current research.
Expensive and time consuming to collect.	Less expensive and saves time as data already exists.
Requires ethical considerations such as informed consent.	Data is in the public domain and has fewer ethical concerns.
Researcher has control over data and accuracy.	Researcher has no control over data accuracy.

Consider the pros and cons of **collecting data yourself (primary)** vs. **getting it from somewhere else (secondary)**

2. Use public / open-source data



However: be wary of *synthetic* (artificially-generated) datasets!

2. Use public / open-source data

FiveThirtyEight

abcNEWS

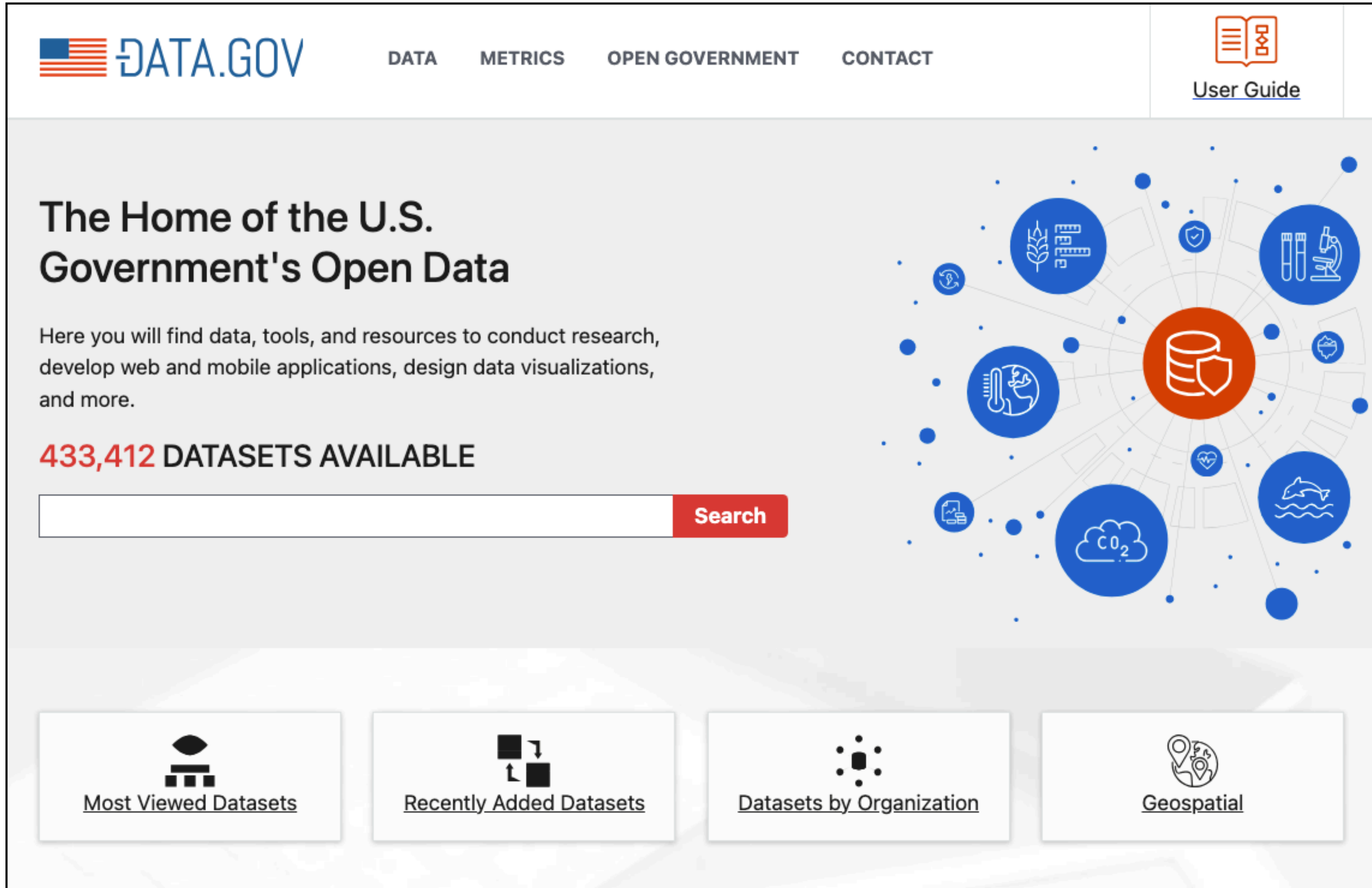
PoliticsSportsScience & HealthEconomicsCulture

Our Data

We're sharing the data and code behind some of our articles and graphics. We hope you'll use it to check our work and to create stories and visualizations of your own.

DATA SET	RELATED CONTENT	LAST UPDATED		
polls	Latest Polls	Sept. 26, 2023	info	↓
congress-generic-ballot	Do Voters Want Democrats Or Republicans In Congress?	Sept. 25, 2023	info	↓
mlb-elo	2023 MLB Predictions	June 20, 2023	info	↓
nba-forecasts	2022-23 NBA Predictions	June 14, 2023	info	↓

2. Use public / open-source data



The screenshot shows the Data.gov homepage. At the top is a navigation bar with the Data.gov logo (an American flag icon followed by 'DATA.GOV') on the left, and links for 'DATA', 'METRICS', 'OPEN GOVERNMENT', and 'CONTACT' in the center. On the right of the navigation bar is a 'User Guide' link with a book icon. Below the navigation bar, the main content area has a light gray background. On the left side of this area, the text 'The Home of the U.S. Government's Open Data' is displayed in a large, bold, black font. Below this, a smaller line of text reads: 'Here you will find data, tools, and resources to conduct research, develop web and mobile applications, design data visualizations, and more.' Further down, the text '433,412 DATASETS AVAILABLE' is shown in a bold, black font, with '433,412' in red. Below this text is a search bar consisting of a white input field and a red 'Search' button. On the right side of the main content area, there is a large, circular graphic composed of many small blue dots connected by thin lines, forming a network. In the center of this network is a large orange circle containing a white icon of a database cylinder with a shield. Surrounding this central circle are several other blue circles, each containing a white icon representing different data categories: a bar chart, a microscope, a globe with a thermometer, a cloud with 'CO2', a fish, a heart, a document, and a leaf. At the bottom of the page, there are four white rectangular buttons with rounded corners, each containing an icon and a text label: 'Most Viewed Datasets' (eye icon), 'Recently Added Datasets' (arrows icon), 'Datasets by Organization' (dots icon), and 'Geospatial' (location pin icon).

DATA.GOV DATA METRICS OPEN GOVERNMENT CONTACT

[User Guide](#)

The Home of the U.S. Government's Open Data

Here you will find data, tools, and resources to conduct research, develop web and mobile applications, design data visualizations, and more.

433,412 DATASETS AVAILABLE

Search

Most Viewed Datasets

Recently Added Datasets

Datasets by Organization

Geospatial

Where to find data for your group project

COGS 9

Home

Syllabus

Readings

Assignments

Group Project

Forming Groups

Project Checkpoint 1

Project Checkpoint 2

Project Checkpoint 2

Video

Advice

Datasets

Exam

Staff

Search COGS 9

Canvas

Piazza

Gradescope

Podcast

Feedback

Group Project

Datasets

Finding Datasets

For the final group project, you will need to find a dataset for your data science question. Here is a list that we have compiled over the years to help you get started. Please feel free to use any of these, or find another dataset online.

NOTE

A few sites may require a free account to download/access the data

- [US Public Food Assistance](#)
This dataset focuses on public assistance programs in the United States that provide food, namely SNAP and WIC
- [DSNY Monthly Tonnage Data](#)
New York's Department of Sanitation dataset about trash and recycling in the city going back to 1990
- [Las Vegas Strip Dataset](#)
This dataset includes quantitative and categorical features from online reviews from 21 hotels located on the Las Vegas Strip, extracted from TripAdvisor. (To view the data set click Data Folder at the top then download the .csv)
- [Bookstore Dataset](#)
Fictional bookstore dataset containing over 1000 books, with different categories, star ratings, and prices
- [Meat Washing Survey Data](#)
YouTuber Adam Ragusea asked his viewers to answer a detailed survey about whether (and why, and how) they wash meat before cooking it. He received more than 13,000 responses, then made a video about what he found and published a spreadsheet of the anonymized answers.

3. Buying data



[Trauma](#) [Quality](#) [National Trauma Data Bank](#)

[Overview](#) [Annual Call For Data](#) [Participant Use File](#) [About](#)

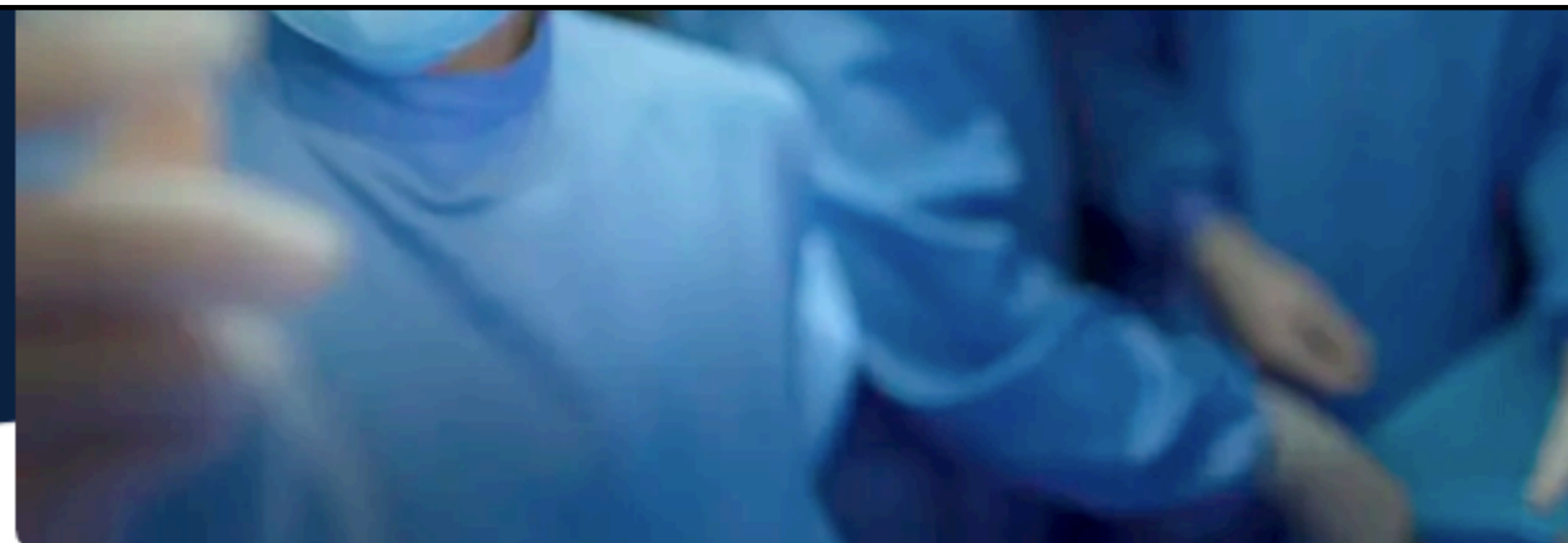
TRAUMA PROGRAM

National Trauma Data Bank® (NTDB®)

Contribute to the largest aggregation of US trauma registry data ever assembled

About the TQP PUF

- The TQP PUF contains all records sent to NTDB for specified admission years
- Includes data submitted on pediatric and adult patients from Level I, II, III, IV, V or undesignated trauma centers
 - Please note that the standard TQP PUF does not contain facility aggregators (a facility-level indicator allowing for the grouping of patients by facility). If you require these data for your project, please request the TQP PUF Facility Key products when completing your TQP PUF application. Additionally, the TQIP participation discount does not apply to TQP PUF Facility Key products.
- The TQP PUF costs \$1,100 per admission year of data requested



3. Buying data

affinitysolutions

Solutions ▾

Industries ▾

Our Data

Our Technology

Newsroom ▾

About Us ▾

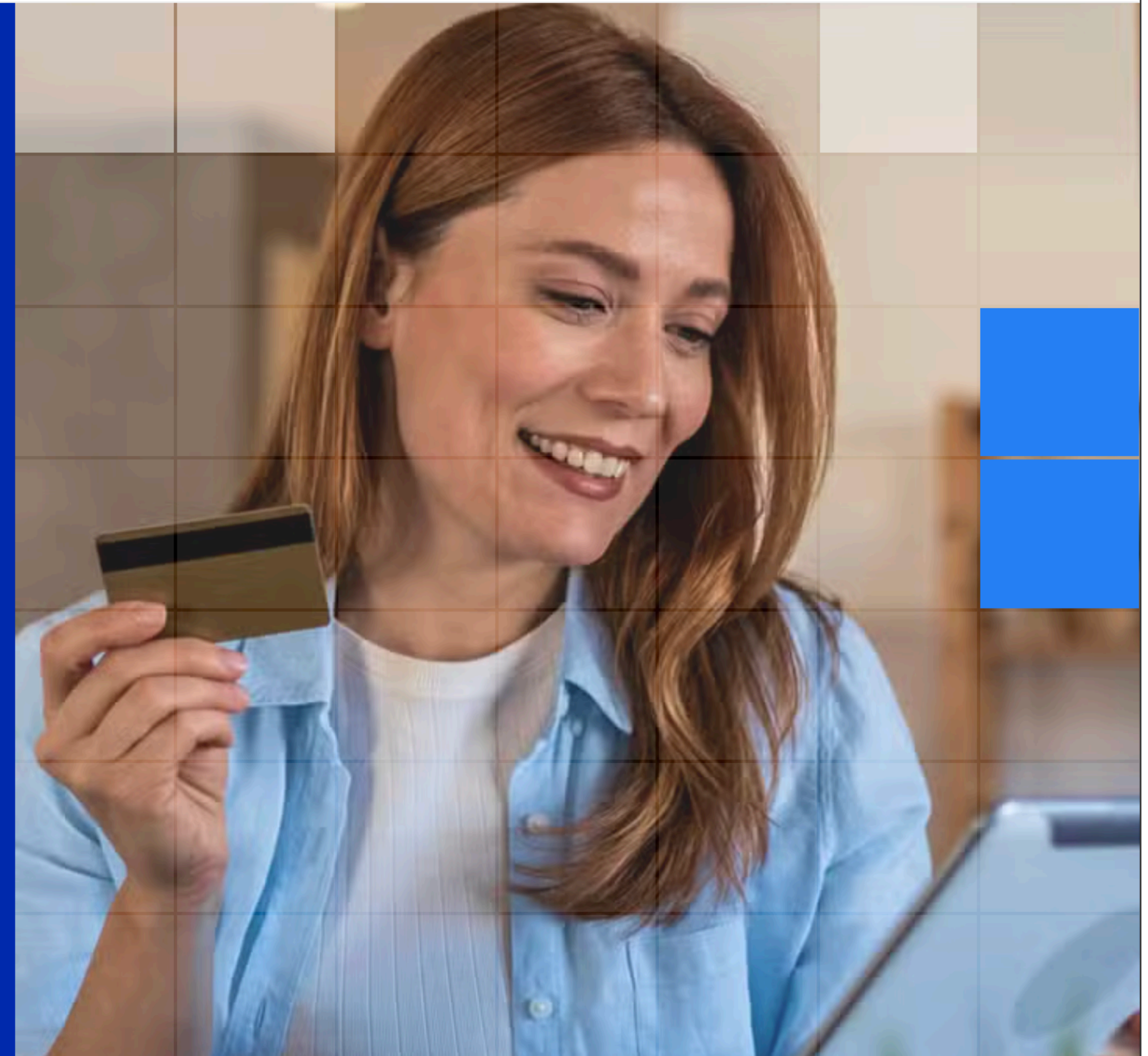
Talk to us

THE LEADING CONSUMER PURCHASE INSIGHTS COMPANY

Consumer Behavior Changes Daily. We Move With It.

Affinity Solutions sees deterministic transaction data from 195M+ consumers in real-time and predicts what they'll buy next, so brands and banks can target smarter, prove impact, continuously optimize, and stay ahead.

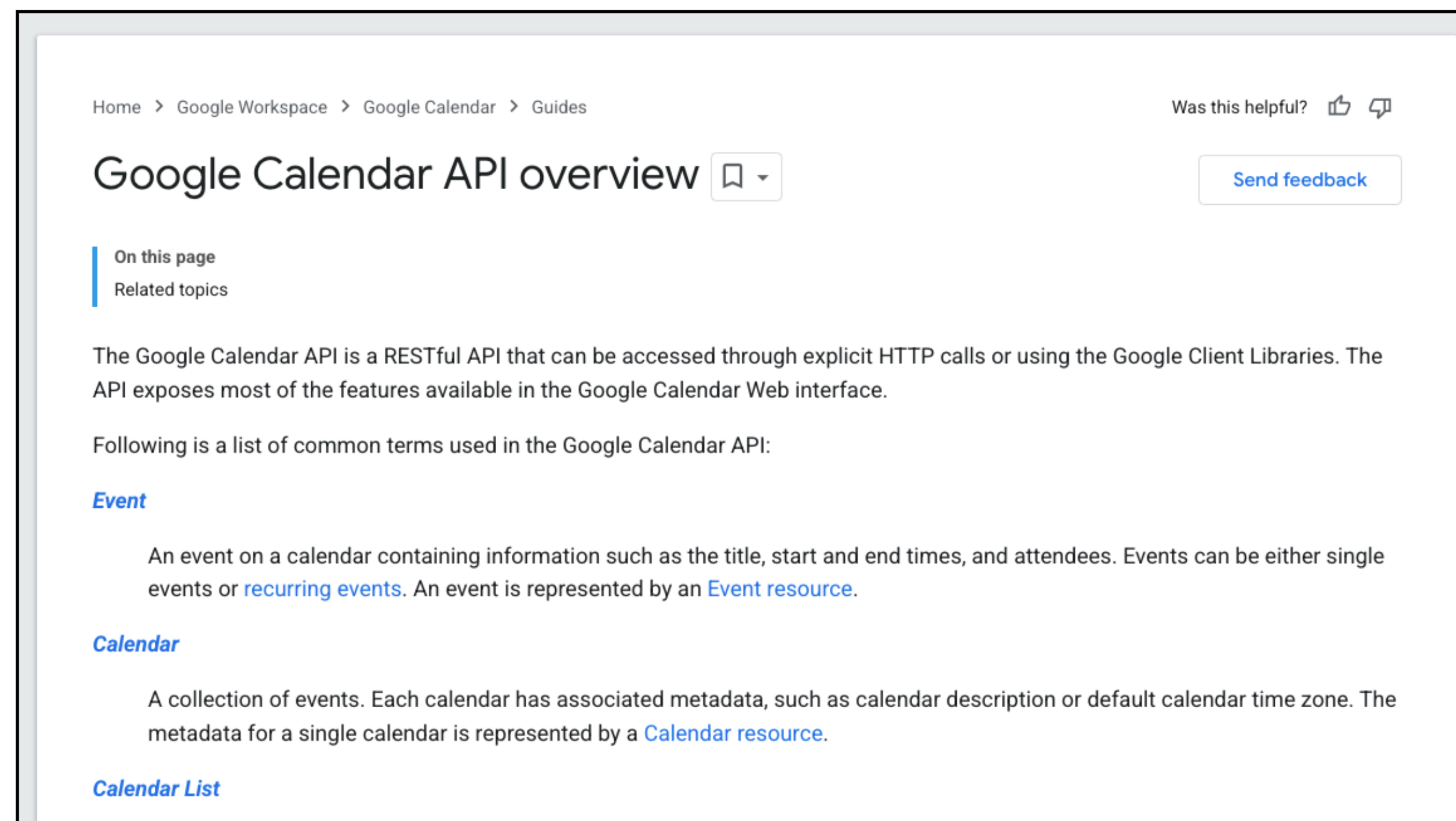
Contact Us →



- Cardlytics (similar company) market cap = ~4 dollars in 2021

4. Requesting it from an API

- API stands for [Application Programming Interface](#)
 - A way for software programs to talk to each other
- **Example:** Where does all of my time go in one week?!...
- I can collect data from my own calendar using the [Google Calendar API](#)



4. Requesting it from an API

Collecting data using an API

1. Send a **request** (e.g., to retrieve my calendar events)
2. API sends back **semi-structured data**, often JSON (e.g. calendar event start times/end times)

```
{
  "summary": "Prep COGS 9 lecture",
  "start": { "dateTime": "2026-02-02T09:00:00-08:00" },
  "end": { "dateTime": "2026-02-02T11:00:00-08:00" }
},
{
  "summary": "Gym",
  "start": { "dateTime": "2026-02-02T17:30:00-08:00" },
  "end": { "dateTime": "2026-02-02T18:30:00-08:00" }
},
{
  "summary": "Meeting with TA team",
  "start": { "dateTime": "2026-02-03T13:00:00-08:00" },
  "end": { "dateTime": "2026-02-03T14:00:00-08:00" }
},
}
```

{.json}

JSON (JavaScript Object Notation)

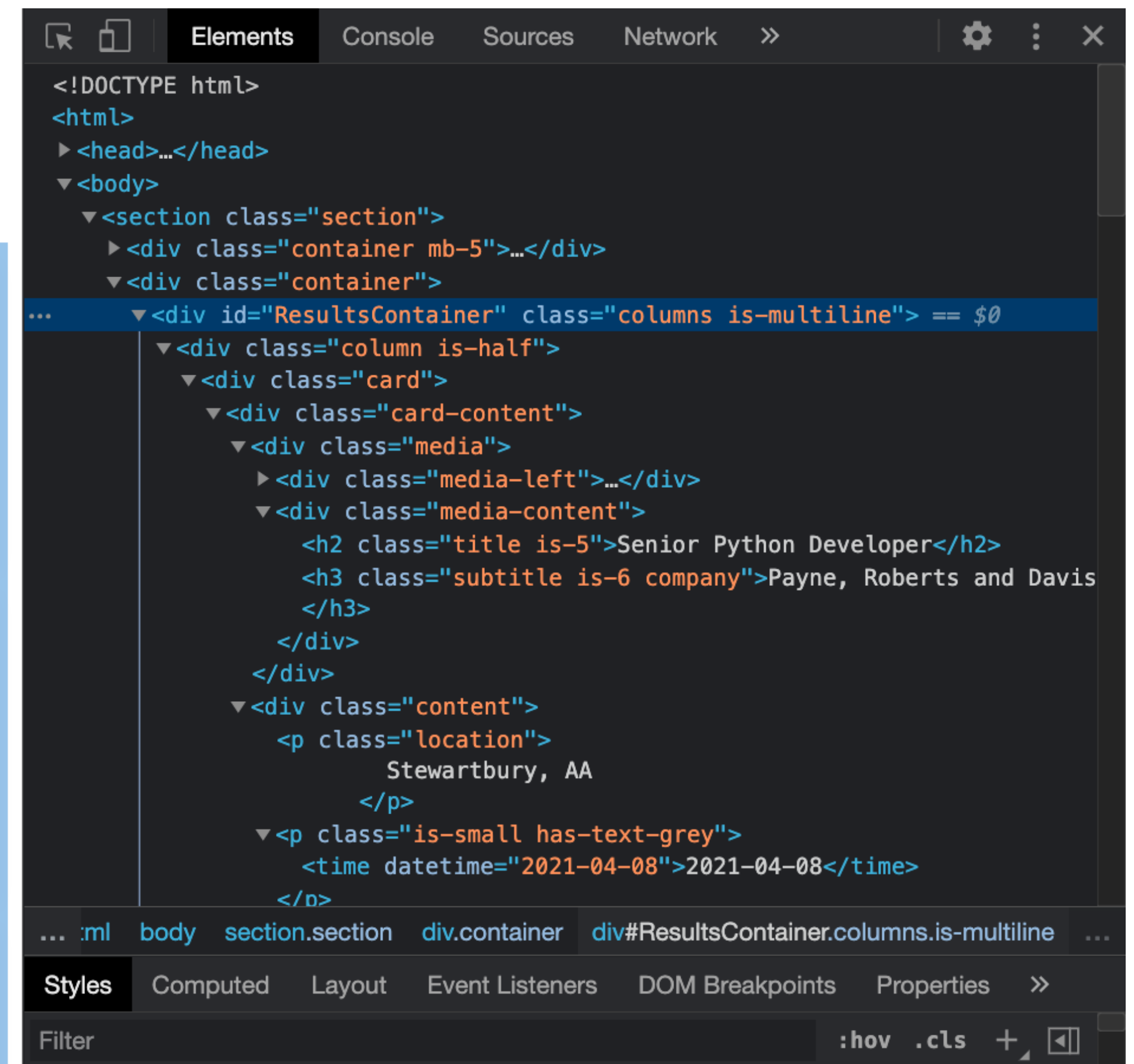
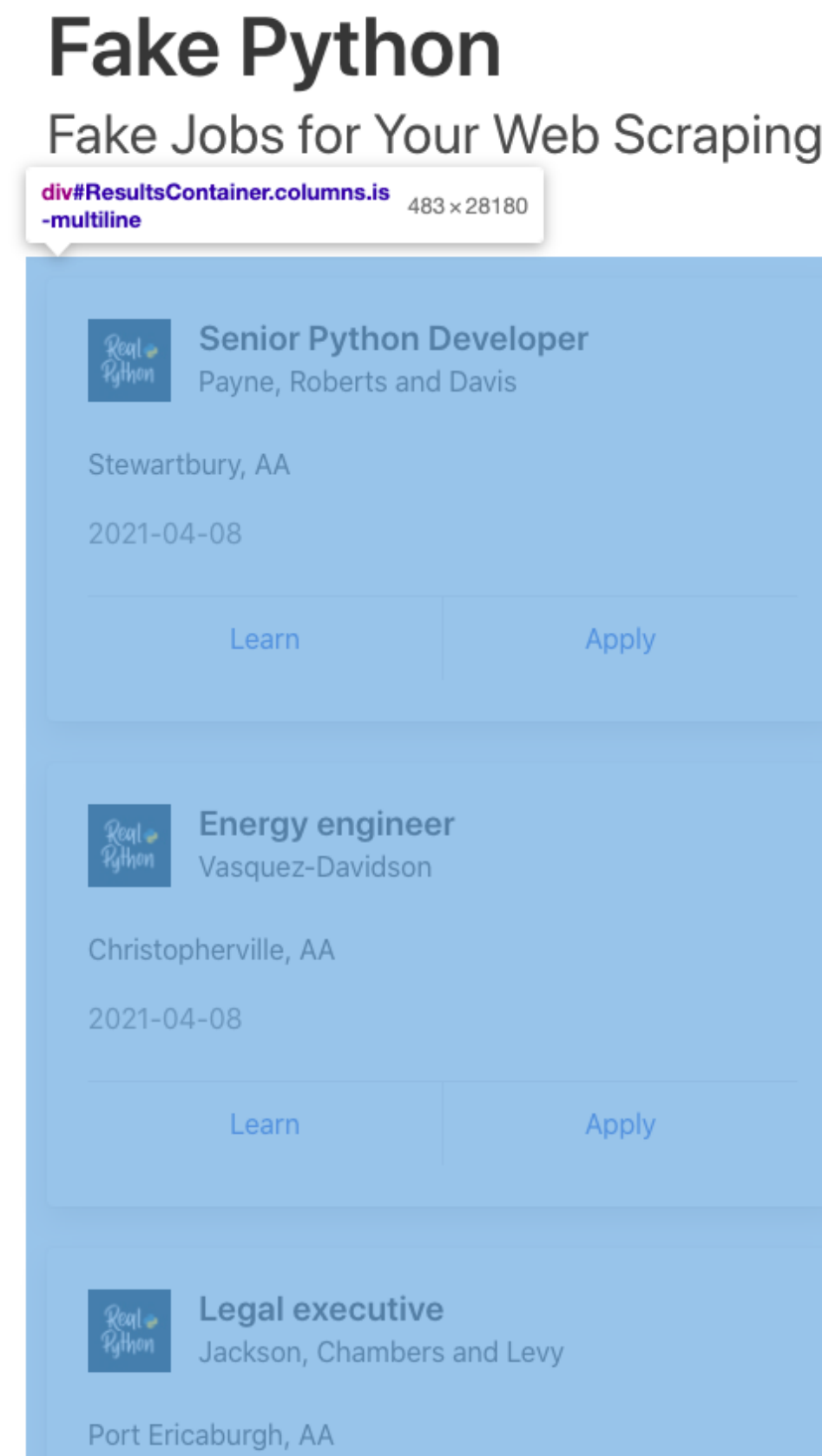
4. Requesting it from an API

Common APIs

- **Maps & locations:** Google Maps
- **Weather:** OpenWeather, WeatherAPI, NOAA (US)
- **Video & music:** YouTube Data API, Spotify Web API
- **Payments:** Stripe API, PayPal API
- **Auth / login:** “Sign in with Google” (Google OAuth)

5. Web scraping

- A way to automatically collect information from websites
- You write a script that downloads a webpage and extracts specific information from the raw HTML code
- Used when there's no API or the API doesn't provide the data you need
- **Example:** You want to build a dataset of entry-level data science jobs



5. Web scraping

- **Example:** You want to build a dataset of entry-level data science jobs
 - Check job description for degree requirements (e.g., no “Master’s degree”)
 - When was the job posted?
 - Find the link to apply

HTML

```
<!-- ... -->
<footer class="card-footer">
  <a href="https://www.realpython.com" target="_blank"
    class="card-footer-item">Learn</a>
  <a href="https://realpython.github.io/fake-jobs/jobs/senior-python-dev"
    target="_blank"
    class="card-footer-item">Apply</a>
</footer>
</div>
</div>
```

Python

```
>>> for job_card in python_job_cards:
...     links = job_card.find_all("a")
...     for link in links:
...         link_url = link["href"]
...         print(f"Apply here: {link_url}\n")
... 
```

Python

scraper.py

```
import requests
from bs4 import BeautifulSoup

URL = "https://realpython.github.io/fake-jobs/"
page = requests.get(URL)

soup = BeautifulSoup(page.content, "html.parser")
results = soup.find(id="ResultsContainer")

python_jobs = results.find_all(
    "h2", string=lambda text: "python" in text.lower()
)

python_job_cards = [
    h2_element.parent.parent.parent for h2_element in python_jobs
]

for job_card in python_job_cards:
    title_element = job_card.find("h2", class_="title")
    company_element = job_card.find("h3", class_="company")
    location_element = job_card.find("p", class_="location")
    print(title_element.text.strip())
    print(company_element.text.strip())
    print(location_element.text.strip())
    link_url = job_card.find_all("a")[1]["href"]
    print(f"Apply here: {link_url}\n")
```

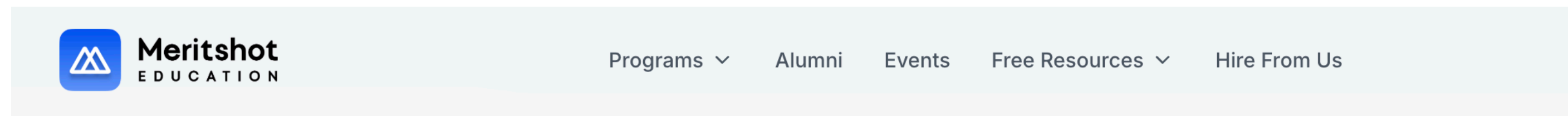
Web scraping vs. APIs

- **APIs:** a site's official way to access data
 - structured and stable
 - may require a key & have rate limits
- **Web scraping:** useful when there's no API
 - more fragile (websites change)
 - can violate terms & conditions
- **Takeaway:** Use an API when available; scrape only when necessary and responsibly!

Criteria	Web Scraping	API
Availability	High availability, as web scraping can be applied to virtually any website	Limited availability, with less than 1% of websites offering an API
Data Attributes	Highly flexible, allowing the user to define custom data extraction parameters for publicly available data	Limited control, with the API's structure and data attributes dictated by the provider
Stability	Moderately stable, with reliance on the consistent structure of targeted websites	Generally high stability, as APIs are maintained and controlled by the providers
Technical difficulty	Requires significant technical expertise, especially for complex websites	Relatively easy to use with basic programming skills, simplifying data access
Cost	Cost varies based on different factors; more cost-effective at larger scales	Costs are predictable, though they can escalate significantly at very large scales
Data Quality	Necessitates robust data cleaning systems to ensure quality	Typically offers structured data, often presented in JSON format

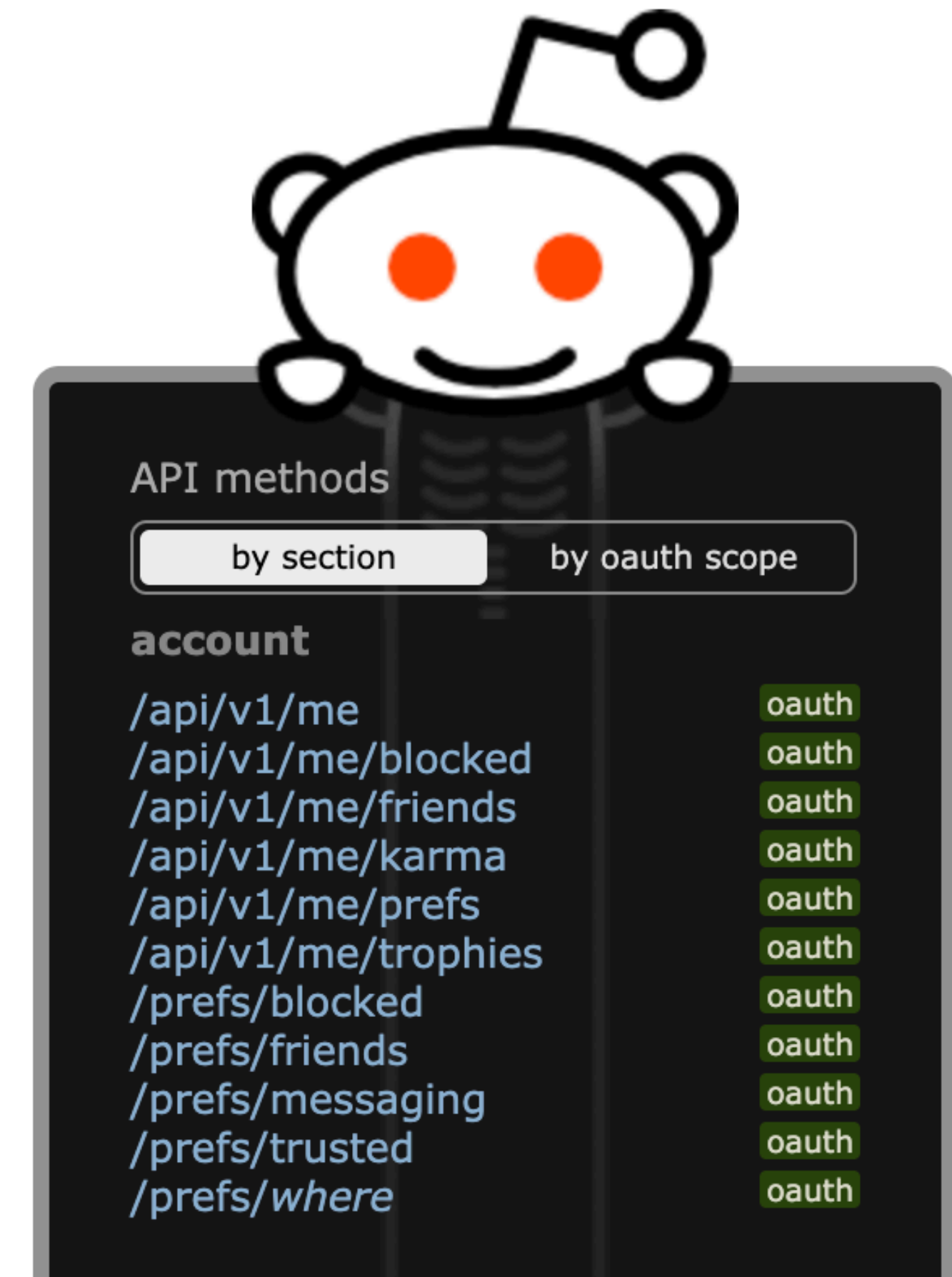
Webscraping, APIs and LLMs

- **Example: Reddit**
 - Pre-LLMs: Free API, no need to web scrape
 - Post-LLMs: Expensive API, web scraping blocked



Reddit Case Study — The Great API Blackout and How It Led to a \$6.4 Billion IPO

AI data licensing emerged as the business model that justified the pricing decision. In early 2024, Reddit announced a data licensing agreement with **Google** valued at approximately **\$60 million annually**



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course code:
WSTAPX

Your project group wants to study the factors that affect US unemployment trends over the last 20 years. What is the best way to obtain this data?

- (A) Collect it yourself
- (B) Buy it
- (C) Use public / open-source data
- (D) Request it from an API
- (E) I'm not sure / I'm confused

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